

Agent-Based Evolutionary Dynamics for Mixed Autonomy Weaving Ramps

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Abstract:

Existing models of mixed-autonomy weaving ramps characterize how altruistic connected and automated vehicles (CAVs) can improve traffic efficiency at the population level, but provide limited insight into how such behavior emerges from decentralized vehicle interactions or how it is affected by finite populations, heterogeneous preferences, and imperfect information. We develop an agent-based model of a macroscopic weaving-ramp framework in which individual vehicles adapt their lane choices using an evolutionary game-theoretic update rule and altruism based objectives providing a microscopic interpretation of the original Wardrop model. We prove convergence of the decentralized dynamics to the unique equilibrium predicted by the macroscopic theory. Beyond reproducing aggregate equilibrium behavior, the framework enables the study of deployment-level questions that cannot be addressed by static analysis. Simulation results demonstrate close agreement with the macroscopic predictions while revealing how convergence rates, adaptation to changing traffic conditions, heterogeneous altruism levels among CAVs, and imperfect state information influence system performance and the distribution of altruistic burden across vehicles. These results provide a bridge between equilibrium traffic theory and decentralized mixed-autonomy deployment.

Keywords: mixed autonomy, weaving ramps, lane choice, evolutionary game theory, agent-based model, social value orientation.

1. INTRODUCTION

Freeway weaving sections, where entering and exiting flows must cross within a short roadway segment, remain one of the most persistent bottlenecks on highways [2]. Recent advances in connected and automated vehicles (CAVs) create an opportunity to influence traffic conditions through lane-choice behavior [14]. Unlike human-driven vehicles (HDVs), whose decisions emerge from individual preferences, CAVs can be programmed to account for broader traffic objectives while remaining decentralized.

Vehicle-level traffic models examine how individual interactions and control decisions affect lane changing, stability, and throughput, while Social Value Orientation (SVO)-based models describe how CAVs balance private and collective costs [15, 12]. A complementary evolutionary-game-theoretic literature studies how strategy shares change over time, from evolutionarily stable strategies and replicator dynamics to traffic-assignment processes that converge to Wardrop equilibria [7, 16, 13].

At the macroscopic scale, recent work has developed a game-theoretic model of lane choice at weaving ramps [4, 18]. The model represents HDVs through a Wardrop equilibrium and incorporates SVO into CAV objectives, allowing CAVs to trade off private delay against aggregate congestion. It characterizes when altruistic CAVs improve traffic efficiency, derives closed-form penetration thresholds, and quantifies the achievable reduction in so-

cial delay. Thus, the mixed-autonomy weaving problem is analytically characterized at equilibrium.

The two modeling perspectives nevertheless leave a complementary gap. The macroscopic theory characterizes equilibrium lane-choice fractions but does not provide a decentralized vehicle-level mechanism for reaching them. Equilibrium existence also does not guarantee convergence under local learning; adjustment may be slow, oscillatory, or unsuccessful [3, 11]. Moreover, the static formulation cannot capture adaptation to changing demand, determine whether finite heterogeneous fleets reproduce mean-field predictions, or explain how the altruistic burden is distributed across vehicles. It also assumes exact traffic information, whereas deployed CAVs rely on noisy local observations and imperfect state estimates. On the other hand, the aforementioned vehicle-level models were not designed to reproduce the calibrated cost structure and penetration thresholds of such a weaving-ramp model that incorporates CAVs as well as HDVs. They therefore do not establish whether finite-agent adaptation recovers its aggregate predictions under realistic deployment constraints.

These limitations motivate the central question: *Can the equilibrium prescriptions of the macroscopic theory be realized by a finite population of decentralized vehicles, and if so, what behavior emerges under adaptation, heterogeneous altruism, and imperfect information?* This paper answers the question by treating the calibrated macroscopic model of [4, 18] as a specification and developing a new agent-

based model (ABM) of it. The equilibrium cost structure and penetration benchmarks are inherited from those papers; the vehicle-level dynamics, convergence analysis, and deployment studies below are novel.

Contributions: (i) We develop a decentralized realization of the macroscopic weaving-ramp model using replicator-based adaptation from evolutionary game theory [20, 11]. We show that the resulting lane-choice game is an exact potential game and that the induced dynamics converge globally to the unique Wardrop equilibrium. (ii) Using the resulting agent-based model, we characterize deployment phenomena that are absent from the equilibrium theory, including convergence regimes, demand-tracking limits, and the distribution of costs and benefits across altruistic and selfish vehicles. (iii) We extend the framework to heterogeneous altruism and partial observability and identify when aggregate behavior is summarized by an effective altruism angle and analyze the impact of noise. (iv) We show that belief sharing improves performance under stationary demand, but can be conservative under abrupt changes. To address this, we introduce a forgetting-factor update that balances accuracy and adaptability.

2. PRELIMINARIES

2.1 Population-game Model

We briefly introduce the population-game concepts used in the paper. Consider a population of agents choosing between two pure strategies, S (stay) and B (bypass), and let $x \in [0, 1]$ denote the fraction of agents selecting S , so that $1 - x$ select B .

A two-strategy *population game* is defined by a pair of cost functions $J^S, J^B : [0, 1] \rightarrow \mathbb{R}$, where $J^S(x)$ and $J^B(x)$ denote the costs incurred by agents choosing strategies S and B , respectively, at population state x . The mean population cost is $\bar{J}(x) = xJ^S(x) + (1 - x)J^B(x)$.

A state $x^* \in [0, 1]$ is a *Wardrop equilibrium* if no agent can reduce its cost by unilaterally switching strategy: $x^* \cdot (J^S(x^*) - J^B(x^*)) \leq 0$; $(1 - x^*) \cdot (J^B(x^*) - J^S(x^*)) \leq 0$. For an interior equilibrium $x^* \in (0, 1)$, this reduces to $J^S(x^*) = J^B(x^*)$. Wardrop equilibrium is the user-equilibrium concept from transportation theory and coincides with the symmetric Nash equilibrium of the corresponding population game [19, 11].

The continuous-time replicator dynamics associated with a population game (J^S, J^B) are

$$\dot{x} = x(1 - x)(J^B(x) - J^S(x)). \quad (1)$$

Replicator dynamics model decentralized adaptation: strategies with lower cost increase their population share over time. Any interior stationary point satisfies $J^S(x^*) = J^B(x^*)$ and therefore coincides with an interior Wardrop equilibrium. In the agent-based model, vehicles update their lane-choice probabilities using a discrete-time replicator rule; as population size grows and the step size shrinks, the stochastic process converges to Eq. (1) [1, 11], so the replicator ODE serves as its analytical approximation.

A population game is an *exact potential game* if there exists a continuously differentiable function $\Phi : [0, 1] \rightarrow \mathbb{R}$ satisfying $\frac{d\Phi}{dx} = J^B(x) - J^S(x)$, $\forall x \in [0, 1]$.

In an exact potential game the stationary points of the replicator dynamics coincide with those of the potential, which can be used directly as a Lyapunov function to establish stability [8, 10, 11].

Let x^S and x^B denote the fractions of vehicles choosing the stay and bypass strategies, ($x^S + x^B = 1$). The agent's private costs are

$$J^{\text{self}, S} = K_1^S x^S + B_1^S, \quad J^{\text{self}, B} = K_1^B x^B + B_1^B \quad (2)$$

The K_1 and B_1 coefficients capture the calibrated traversing and merging costs; their detailed expressions are provided in Section 2.2. The aggregate social cost is the sum of the costs incurred by all traffic classes, weighted by their respective flow fractions and is denoted by J^{soc} .

Social Value Orientation (SVO) represents an agent's relative concern for private and collective outcomes by an altruism angle $\theta \in [0, \pi/2]$. For a lane choice $a \in S, B$, the agent minimizes the effective cost

$$\tilde{J}^a(\theta) = \cos(\theta) J^{\text{self}, a} + \sin(\theta) (\partial J^{\text{soc}} / \partial x^a), \quad (3)$$

where $J^{\text{self}, a}$ is the agent's private travel cost and $\frac{\partial J^{\text{soc}}}{\partial x^a}$ is the marginal social cost of its lane choice, and the sine-cosine weights are the standard unit-circle parameterization of SVO [17, 9, 6]. The boundary cases are $\theta = 0$ (fully selfish) and $\theta = \pi/2$ (fully altruistic).

2.2 Weaving Ramp Model

We now present the macroscopic weaving-ramp model [4, 18] and state equilibrium predictions that the agent-based model of Sec. 3 will be shown to match.

Setup and Parameters. We consider a single highway weaving section with two lanes. Vehicles are partitioned into three movement classes by their origin and destination: *entering* vehicles, which join the highway at the on-ramp; *exiting* vehicles, which leave at the off-ramp; and *through* vehicles, which traverse the section without changing lanes. Each *through* vehicle in the mainline lane adjacent to the ramps additionally selects one of two lane-choice strategies: *stay* (S), remaining in lane 1, or *bypass* (B), shifting to lane 2 to avoid the merge conflict. The model is one-shot at the equilibrium-flow scale: lane choices within the weaving region are resolved into stationary flow ratios, and intra-section spatial dynamics are abstracted into the lane-cost functions below.

Let $n = (n_{\text{enter}}, n_{\text{exit}}, n_2)$ denote the normalized demand composition, where they denote the fractions of entering, exiting, and through vehicles, respectively. Let $x \in [0, 1]$ denote the fraction of lane 1 through vehicles choosing S , so that $1 - x$ choose B . Let $p \in [0, 1]$ denote the CAV penetration rate; the remaining $1 - p$ fraction are HDVs. The cost functions depend on four coefficients ($C_1^t, C_1^m, C_2^t, C_2^m$) from optimization method in [5] and six coefficients ($\alpha, \beta, \gamma, \delta, \omega, \rho$), calibrated against microsimulation in [4].^{1 2}

Lane Costs. The travel delay costs of S and B are affine in the lane-choice state x , as seen in (2): with $K_1^S = C_1^t \alpha + C_1^m (\omega n_{\text{exit}} + n_{\text{enter}})$, $B_1^S = C_1^t (\beta n_{\text{exit}} + n_{\text{enter}})$, $K_1^B = C_2^t \gamma + C_2^m (\rho n_2 + \delta n_{\text{exit}})$, $B_1^B = C_2^t n_2$.

¹ ($C_1^t, C_1^m, C_2^t, C_2^m$) = (1.000, 1.000, 1.000, 1.000)

² ($\alpha, \beta, \gamma, \delta, \omega, \rho$) = (1.255, 1.138, 2.384, 3.094, 1.000, 1.000).

The aggregate social cost $J^{\text{soc}}(x, n)$ is the total delay experienced by all vehicles in the weaving section, obtained by summing each class's individual cost weighted by its flow: (i) steadfast and bypassing vehicles in lane 1, (ii) through vehicles in lane 2, and (iii) the entering and exiting vehicles and can be given as: $J^{\text{soc}} = x^S(J^{\text{self},s}) + x^B(J^{\text{self},s}) + n_2(K_2^S x^B + B_2^S) + n_{\text{exit}}(K_2^{\text{exit}} x^S + B_2^{\text{exit}}) + n_{\text{enter}}(K_0^{\text{enter}} x^S + B_0^{\text{enter}})$. The strategy-specific marginal social costs are

$$M^a(x, n) := \partial J^{\text{soc}} / \partial x^a \big|_{(x^S, x^B) = (x, 1-x)}, \quad a \in \{S, B\}. \quad (4)$$

For the calibrated affine lane-cost model, J^{soc} is quadratic in x , and M^s and M^b are affine.

Equilibrium Benchmarks. Two benchmark lane-choice states organize the analysis (i) *HDV Wardrop point* $\xi(n)$, the equilibrium of the pure-HDV population ($p = 0$), obtained by setting $J_1^S = J_1^B$ in (2), and (ii) the *social optimum* $\Gamma(n)$, the steadfast fraction that minimizes the aggregate cost:

$$\xi(n) = \frac{K_1^B + B_1^B - B_1^S}{K_1^S + K_1^B}; \quad \Gamma(n) = \arg \min_{x \in [0,1]} J^{\text{soc}}(x, n). \quad (5)$$

The admissible configurations of interest are those in which the selfish equilibrium and the social optimum are distinct, i.e. $0 < \xi(n) < \Gamma(n) < 1$; under this condition altruistic CAVs can steer the population from ξ toward Γ .³

Three-Phase Penetration Structure. The aggregate impact of altruistic CAVs as the CAV penetration p varies is established in [18]: for any admissible configuration, J^{soc} at the population equilibrium follows a three-phase structure in p . (i) At low penetration $p \in [0, p_1]$, J^{soc} remains at the HDV-only reference level; (ii) over an intermediate band $p \in (p_1, p_2]$, it decreases strictly in p ; (iii) and for $p \in (p_2, 1]$, it saturates at the social-optimum level.⁴ This three-phase structure is the central macroscopic prediction that the ABM of Sec.3 will be shown to reproduce.

SVO Effective Cost. Specializing the definition of SVO (3) to the present setting, a CAV with altruism angle $\theta \in [0, \pi/2]$ minimizes the effective cost:

$$\tilde{J}^{(\cdot)}(x, n; \theta) = \cos(\theta) J_1^{(\cdot)}(x) + \sin(\theta) M^{(\cdot)}(x, n), \quad (6)$$

for $(\cdot) \in \{S, B\}$, with $J_1^{(\cdot)}$ given by (2) and M by (4). The angle θ is a single broadcastable parameter: HDVs use $\theta = 0$, while the CAV fleet adopts a value assigned by the operator. The induced mixed-population equilibrium interpolates between ξ (at $\theta = 0$) and Γ (at $\theta = \pi/2$), and is the object the AVM of the next section converges to.

3. AGENT-BASED MODEL AND THEORY

This section constructs the agent-based model (ABM) that realizes the aggregate statements over population fractions of the macroscopic model at the level of individual vehicles.

³ This ordering applies to operational regimes in which selfish traffic supplies fewer steadfast vehicles and increasing the weight on marginal social cost moves the equilibrium from ξ toward Γ .

⁴ The two transition points are: $p_1 = \frac{K_1^B + B_1^B - B_1^S}{K_1^S + K_1^B}$, and $p_2 = \frac{(2K_1^B + B_1^B - B_1^S - n_{\text{exit}}K_2^{\text{exit}} - n_{\text{enter}}K_0^{\text{enter}} + n_2K_2^S)}{(2(K_1^S + K_1^B))}$ where K_2^{exit} and K_0^{enter} are the x^S -coefficients in exiting and entering costs, while K_2^S is the x^B -coefficient in Lane-2 through costs.

The ABM uses the lane costs (2), the social-cost machinery, and the SVO effective cost (6) as specified in Sec. 2.2.

Agents and Lane-Choice Dynamics. We consider a population of N vehicles indexed by $i \in \{1, \dots, N\}$. Each vehicle is characterized by its (fixed) movement class $c_i \in \{\text{enter}, \text{exit}, \text{through}\}$, its (fixed) altruism angle $\theta_i \in [0, \pi/2]$, and its (time-varying) lane-choice probability $\pi_i(t) \in [0, 1]$ for selecting the steadfast strategy S .

HDVs have $\theta_i = 0$; CAVs have $\theta_i > 0$, with the value broadcast by the fleet operator.

The aggregate steadfast fraction is given by (7). Each vehicle updates its lane-choice probability by a discrete-time replicator rule applied to its own SVO cost:

$$\pi_i(t+1) = \pi_i(t) + \eta \pi_i(t)(1 - \pi_i(t))[\tilde{J}_i^B(x(t)) - \tilde{J}_i^S(x(t))] \quad (8)$$

where $\tilde{J}_i^{(\cdot)}$ is the effective cost (6) evaluated at the agent's own angle θ_i , and $\eta > 0$ is a step size.⁵ The rule is decentralized: each vehicle uses its own altruism angle, the realized lane costs, and the demand composition needed to form the marginal social cost M . The operator's only intervention is the choice of θ broadcast to the CAV fleet.

Mean-Field Approximation and Convergence.

We derive a scalar continuous-time approximation using the rescaled time $\tau = \eta t$. This follows standard stochastic-approximation arguments [1], and its agreement with the aggregate agent-based dynamics is assessed numerically.

Let $\Delta_0(x) := J_1^B(x) - J_1^S(x)$ denote the cost advantage of choosing the steadfast strategy for an HDV. For a CAV with altruism angle θ , the corresponding effective-cost difference is $\Delta_\theta(x) = \cos \theta \Delta_0(x) + \sin \theta [M^B(x) - M^S(x)]$. Suppose that a fraction $1 - p$ of the strategic vehicles are HDVs and a fraction p are CAVs with SVO angle θ . Their population-average incentive is then:

$$\Delta^{\text{pop}}(x) = (1 - p)\Delta_0(x) + p\Delta_\theta(x) \quad (9)$$

From (7) and (8), the exact aggregate change generated by the individual probability updates is $\frac{x(t+1) - x(t)}{\eta} = \frac{1}{N} \sum_{i=1}^N \pi_i(t)(1 - \pi_i(t))\Delta_{\theta_i}(x(t))$. To obtain a scalar approximation, we assume that the individual probabilities remain concentrated around their population average, giving: $(1/N) \sum_{i=1}^N \pi_i(t)(1 - \pi_i(t))\Delta_{\theta_i}(x) \approx x(1 - x)\Delta^{\text{pop}}(x)$.

$\dot{x} = x(1 - x)\Delta^{\text{pop}}(x)$, (10) After introducing the rescaled time $\tau = \eta t$, the resulting continuous-time approximation is (10). Intuitively, $x(1 - x)$ measures the population's responsiveness, whereas $\Delta^{\text{pop}}(x)$ measures its average incentive to choose the steadfast strategy.

Theorem 1. (Aggregate potential and convergence). Under the homogenization closure, suppose $K^S + K^B > 0$ and $a > 0$.⁶ Then, for every $p \in [0, 1]$ and $\theta \in [0, \pi/2]$:

(i) Φ^{pop} is a strictly concave exact potential for the aggregate mean-field game and therefore has a unique maximizer $x^* \in [0, 1]$.

(ii) If $x^* \in (0, 1)$, it is the unique interior aggregate

⁵ We assume that η is small enough that $\eta|\tilde{J}_i^B - \tilde{J}_i^S| \leq 1$, ensuring that $\pi_i(t) \in [0, 1]$ is preserved by the update.

⁶ With positive coefficients $(\alpha, \beta, \gamma, \delta, \omega, \rho)$ and nonnegative demand fractions, J_{soc} is strictly convex in x with $a = 4(\alpha + \gamma + n_{\text{enter}} + (\omega + \delta)n_{\text{exit}} + \rho n_2) > 0$, identical for the lane-1 and full social-cost forms since the additional movement terms are linear in x . Likewise $K^S + K^B = \alpha + \gamma + n_{\text{enter}} + (\omega + \delta)n_{\text{exit}} + \rho n_2 > 0$.

Wardrop equilibrium and the unique interior stationary point of (10). It is globally asymptotically stable on $(0, 1)$ with Lyapunov function $V(x) = \Phi^{\text{pop}}(x^*) - \Phi^{\text{pop}}(x)$.

(iii) Under $u = -J$, the interior maximizer is the unique symmetric Nash equilibrium of the representative aggregate game and is evolutionarily stable.⁷

Proof. [Sketch] By definition, $\Phi'_0 = \Delta_0$ and $\Phi'_\theta = \Delta_\theta$, we have $(\Phi^{\text{pop}})' = \Delta^{\text{pop}}$, proving the exact potential property. Moreover, from (9) $(\Phi^{\text{pop}})'' = (\Delta^{\text{pop}})' = -(1-p)(K_1^S + K_1^B) - p[(K_1^S + K_1^B)\cos\theta + a\sin\theta] < 0$, so the potential is strictly concave and has a unique maximizer. When this maximizer is interior, $(\Phi^{\text{pop}})'(x^*) = 0$ is both the aggregate Wardrop condition and the stationary condition for (10). Furthermore, $\dot{V} = -x(1-x)[(\Phi^{\text{pop}})'(x)]^2 < 0$ for every $x \in (0, 1) \setminus \{x^*\}$, proving global asymptotic stability. The Nash and ESS conclusions follow from the strict potential maximum [8, 11, 20].

Heterogeneous Fleets. We next allow each CAV's fixed altruism angle to be drawn from a distribution F on $[0, \pi/2]$. Under the same aggregate homogenization closure, the expected cost difference becomes

$$\Delta_F^{\text{pop}}(x) = (1-p)\Delta_0(x) + p\mathbb{E}_F[\cos\theta]\Delta_0(x) + p\mathbb{E}_F[\sin\theta][M^B(x) - M^S(x)]. \quad (11)$$

Thus, at the aggregate mean-field level, F enters the dynamics only through $\mathbb{E}_F[\cos\theta]$ and $\mathbb{E}_F[\sin\theta]$. Define the effective angle and coherence factor by $\bar{\theta}_{\text{eff}} = \text{atan2}(\mathbb{E}_F[\sin\theta], \mathbb{E}_F[\cos\theta])$, $\rho_F = \sqrt{\mathbb{E}_F[\cos\theta]^2 + \mathbb{E}_F[\sin\theta]^2}$. Then $\mathbb{E}_F[\cos\theta] = \rho_F \cos \bar{\theta}_{\text{eff}}$, $\mathbb{E}_F[\sin\theta] = \rho_F \sin \bar{\theta}_{\text{eff}}$, and (11) can be written as $\Delta_F^{\text{pop}} = (1-p)\Delta_0 + p\rho_F\Delta_{\bar{\theta}_{\text{eff}}}$. Equivalently, let $\lambda_F := 1 - p + p\rho_F$, $p_{\text{eff}} := \frac{p\rho_F}{1-p+p\rho_F}$. The heterogeneous drift then factors as $\Delta_F^{\text{pop}} = \lambda_F[(1-p_{\text{eff}})\Delta_0 + p_{\text{eff}}\Delta_{\bar{\theta}_{\text{eff}}}]$. Hence its equilibrium and phase portrait coincide with those of the homogeneous aggregate model having angle $\bar{\theta}_{\text{eff}}$ and penetration p_{eff} ; the positive factor λ_F changes only the time scale.

Because $\rho_F \leq 1$, with equality if and only if all CAVs share the same altruism angle, heterogeneity reduces the fleet's effective aggregate influence. The conclusions of Thm. 1 therefore continue to hold for the heterogeneous aggregate mean-field model after replacing (p, θ) by $(p_{\text{eff}}, \bar{\theta}_{\text{eff}})$.

This equivalence concerns the expected aggregate dynamics. Finite fleets with persistent agent-specific altruism may differ because lane-choice probabilities can become correlated with behavioral type and because of finite-population sampling variability.

Partial Observability and Cooperative Belief Sharing. The SVO effective cost in (6) depends on the demand composition $n = (n_{\text{enter}}, n_{\text{exit}}, n_2)$, which determines both the private lane costs and the marginal social costs. The macroscopic model assumes that n is known exactly. In practice, a deployed CAV has access only to noisy local observations of the traffic state. We model the measurement of vehicle i at time t as: $y_{i,t} = n + \varepsilon_{i,t}$, $\varepsilon_{i,t} \sim \mathcal{N}(0, \sigma_{\text{obs}}^2 I)$, with observations independent across vehicles and time.

⁷ The model is written in costs (which agents minimize), whereas potential-game and evolutionary-stability statements are conventionally written in payoffs (which agents maximize). We use the payoff convention $u = -J$. Under this, lower-cost strategy corresponds to higher potential, and convexity of social cost to concavity of the potential.

Each vehicle replaces n in (6) by an estimate $\hat{n}_{i,t}$ formed from available information. We compare two estimators: (i) *Memoryless estimator.* The vehicle uses only its most recent observation, $\hat{n}_{i,t}^{\text{mem}} = y_{i,t}$. This models a one-shot lane decision based solely on the current local measurement. (ii) *Cooperative belief-sharing estimator.* Vehicles maintain and communicate beliefs regarding the traffic composition. Let $\mu_{i,t}$ denote vehicle i 's belief and let $\tau_{i,t} = 1/\sigma_{i,t}^2$ denote its precision. Whenever a new observation $y_{j,t+1}$ is received from another vehicle or roadside infrastructure, the belief is updated according to $\tau_{i,t+1} = \tau_{i,t} + \frac{1}{\sigma_{\text{obs}}^2}$, $\mu_{i,t+1} = \mu_{i,t} + \frac{1/\sigma_{\text{obs}}^2}{\tau_{i,t+1}}(y_{j,t+1} - \mu_{i,t})$, and the estimate used in the SVO cost is $\hat{n}_{i,t}^{\text{share}} = \mu_{i,t}$. This update is the standard Bayesian fusion rule for independent Gaussian measurements and can be interpreted as a simple distributed information-sharing protocol. The next result quantifies the benefit of information-sharing: combining independent observations keeps the demand estimate unbiased while reducing variance.

Lemma 2. (Estimator bias and variance). Both estimators are unbiased, $\mathbb{E}[\hat{n}_{i,t}^{\text{mem}}] = \mathbb{E}[\hat{n}_{i,t}^{\text{share}}] = n$, with $\text{Var}[\hat{n}_{i,t}^{\text{mem}}] = \sigma_{\text{obs}}^2 I$ and $\text{Var}[\hat{n}_{i,t}^{\text{share}}] = (\sigma_{\text{obs}}^2/t) I$.

Proof. The memoryless estimate is $\hat{n}_{i,t}^{\text{mem}} = y_{i,t} = n + \varepsilon_{i,t}$, giving mean n and variance $\sigma_{\text{obs}}^2 I$. Bayesian fusion of t independent Gaussian observations has posterior mean equal to the sample average $\hat{n}_{i,t}^{\text{share}} = \frac{1}{t} \sum_{k=1}^t y_{i,k}$, hence mean n and variance $\sigma_{\text{obs}}^2/t I$.

Because the effective-cost difference depends linearly on the estimated demand, both unbiased estimators recover the full-information equilibrium on average. They differ only in how much noise they introduce into the updates.

Corollary 3. (Equilibrium is preserved). The effective-cost difference $\Delta(x, n) = \tilde{J}^B - \tilde{J}^S$ is affine in n , so $\Delta(x, n) = a(x)^\top n + b(x)$ and, by Lemma 2, $\mathbb{E}[\Delta(x, \hat{n}_{i,t}) | x] = a(x)^\top n + b(x) = \Delta(x, n)$ for either estimator. The expected replicator vector field and the equilibrium location are thus identical to the full-information case; the estimators differ only in variance, which is $O(\sigma_{\text{obs}}^2)$ for the memoryless rule and decays as $1/t$ under belief sharing.

4. EXPERIMENTAL RESULTS

We evaluate the agent-based realization of Sec. 3 using the parameters of [18], with $N = 1000$ agents and step size $\eta = 0.08$.

Macroscopic-equilibrium validation. For the HDV-only case ($p = 0$), Fig. 1 compares the ABM steady-state bypass fraction with the closed-form Wardrop equilibrium $\xi(n)$ and an independent SUMO simulation. All three agree within 1–2% across the tested demand conditions.

For mixed traffic, Fig. 2 compares the analytic equilibrium and ABM results over CAV penetration p and altruism angle θ . Both recover the three regimes of p_1 and p_2 : an HDV baseline for $p \leq p_1$, a decreasing transition, and a θ -dependent floor for $p \geq p_2$.

Heterogeneous fleets. We test four homogeneous profiles and five mixtures over $\theta \in \{\pi/6, \pi/4, \pi/3, \pi/2\}$. As shown in Fig. 3, all profiles retain the three-regime struc-

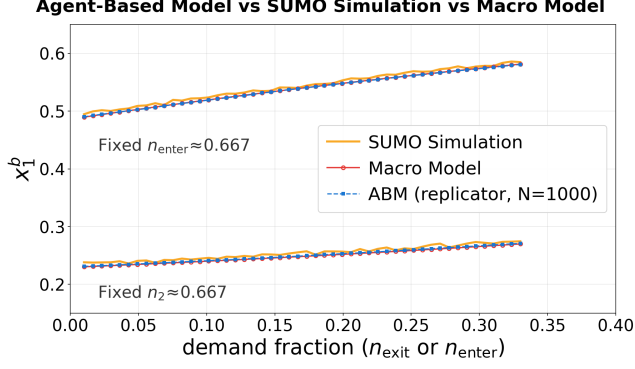


Fig. 1. HDV-only ABM, closed-form Wardrop equilibrium, and independent SUMO simulation.

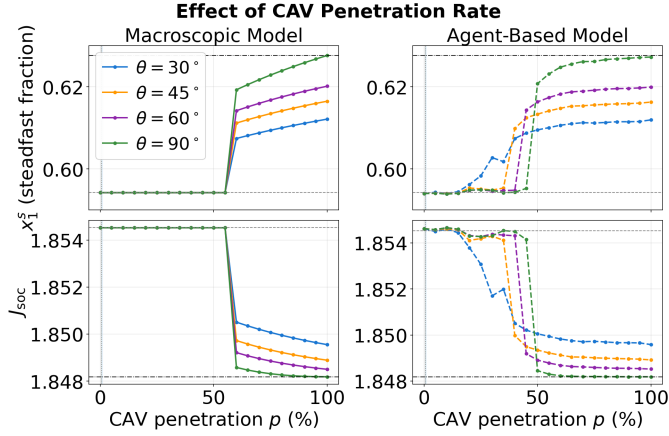


Fig. 2. Analytic mixed-population equilibrium (left) and ABM results (right), including thresholds p_1 and p_2 .

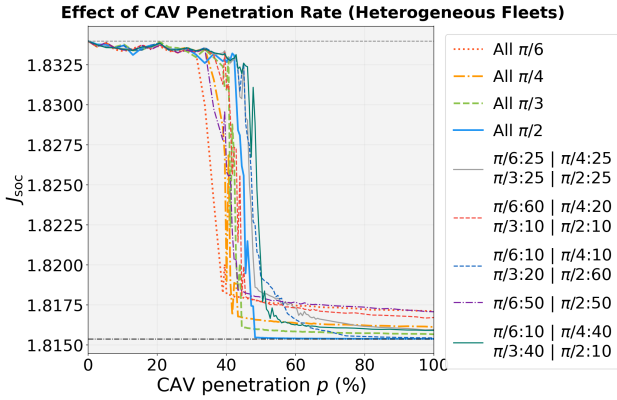


Fig. 3. Social cost for homogeneous and heterogeneous CAV fleets. Profiles with matched $\bar{\theta}_{\text{eff}}$ have similar equilibria.

ture. Profiles with the same $\bar{\theta}_{\text{eff}}$ produce nearly identical equilibria, while greater heterogeneity primarily affects transient convergence through the coherence factor ρ_F .

Convergence speed and stability. Fig. 4 reports the settling time T_{eq} over (p, η_{CAV}) at $\theta = \pi/2$. Low-penetration fleets require conservative learning rates, whereas higher penetration supports faster adaptation without sustained oscillation.

Tracking time-varying demand. We vary $n_2(t)$ sinusoidally and compare the ABM trajectory with the in-

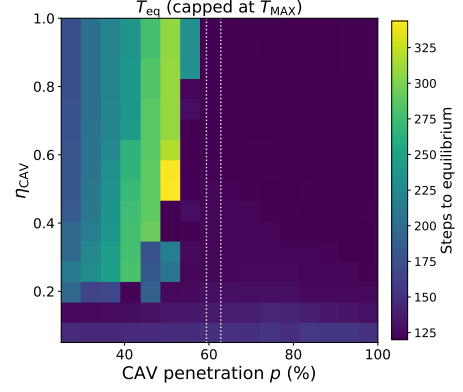


Fig. 4. Steps to equilibrium over (p, η_{CAV}) at $\theta = \pi/2$; dotted lines indicate p_1 and p_2 .

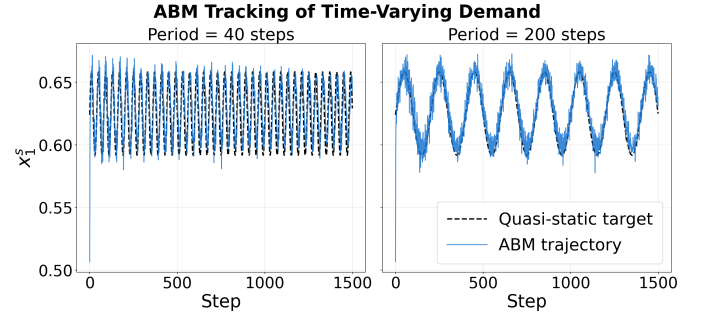


Fig. 5. ABM tracking of sinusoidal demand with periods 40 and 200 for $p > p_2$.

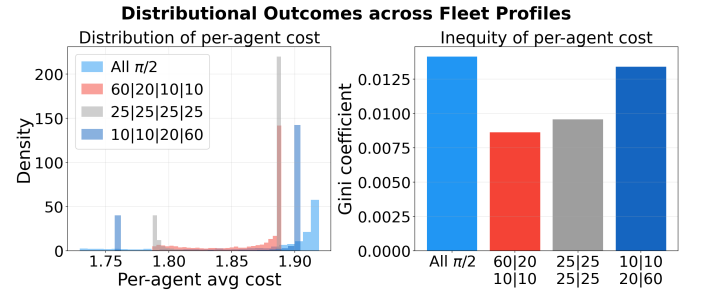


Fig. 6. Per-agent cost distributions (left) and Gini coefficients (right) across CAV fleet profiles.

stantaneous quasi-static equilibrium. Fig. 5 shows that the fleet follows slowly varying demand more accurately: the RMSE is ≈ 0.007 for period 200 and 0.012 for period 40.

Distributional outcomes and free-riding. Aggregate social cost does not reveal how delay is distributed across vehicles. Fig. 6 compares per-agent costs across homogeneous and mixed SVO profiles. The fully altruistic fleet has the highest measured inequity ($G \approx 0.014$), while the mixed profiles have lower inequity ($G \approx 0.009$). Thus, full altruism can concentrate the cooperative burden on the vehicles that absorb the conflict-zone delay.

Partial observability and belief sharing. True demand n is replaced by a noisy estimate $y_{i,t} = n + \varepsilon_{i,t}$, where $\varepsilon_{i,t} \sim \mathcal{N}(0, \sigma_{\text{obs}}^2 I)$. Fig. 7 shows that observation noise delays the transition between penetration regimes but does not change the social-optimum floor. Cooperative belief sharing is closer to the perfect-information result than a memoryless estimator.

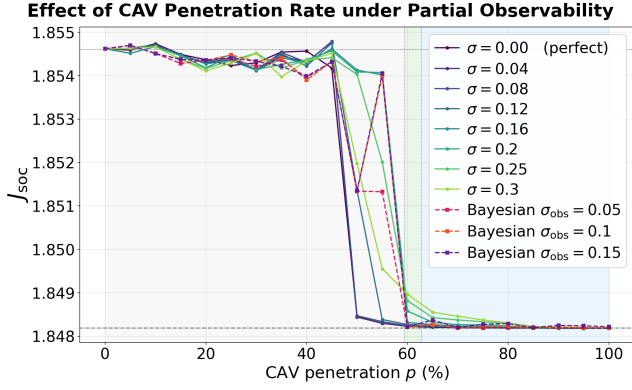


Fig. 7. Social cost under perfect information, memoryless estimation, and cooperative belief sharing.

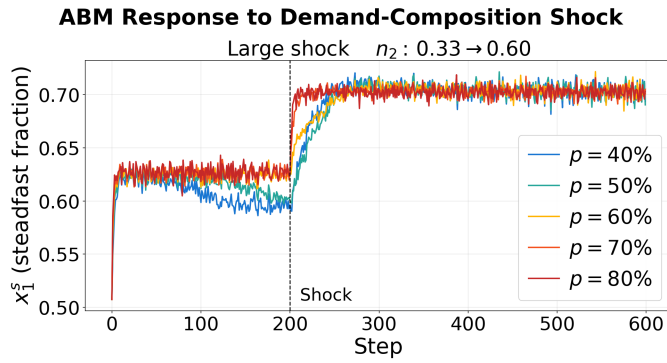


Fig. 8. Recovery after a demand-composition shock across CAV penetration rates.

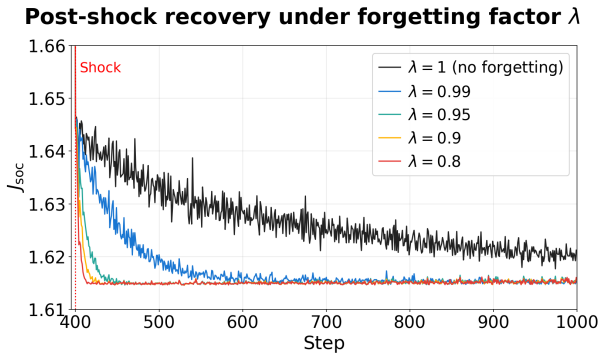


Fig. 9. Post-shock recovery of social cost under different forgetting factors λ ; the shock occurs at $t = 400$.

Recovery from demand shocks. We equilibrate the fleet at balanced demand and apply a shock $n_2 : 0.33 \rightarrow 0.60$ at $t = 200$. As shown in Fig. 8, all fleets reach the same post-shock equilibrium, but recovery is faster at higher penetration: $p \geq 70\%$ settles within a few steps, whereas $p = 40\text{--}50\%$ requires approximately 30–40 steps.

Belief sharing under shocks. Bayesian belief sharing reduces estimation variance under stationary demand but can retain stale information after a shock. We therefore discount past observations using $\tau_{t+1} = \lambda\tau_t + \sigma_{\text{obs}}^{-2}$, $\lambda \in (0, 1]$. Fig. 9 shows that standard fusion ($\lambda = 1$) may require hundreds of steps to recover, whereas $\lambda \leq 0.95$ restores convergence within a few steps.

5. CONCLUSIONS

We developed an agent-based realization of the macroscopic weaving-ramp altruism framework, proved its convergence to the macroscopic equilibrium, and used it to derive deployment-level findings invisible to the static analysis. These results are limited to a single weaving ramp under a calibrated affine-cost model with homogeneous learning rates. Future work includes networks of coupled weaving sections, and heterogeneous and adaptive learning dynamics.

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